

MACHINE LEARNING PROJECT

2025

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AGENDA

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PROJECT DESCRIPTION

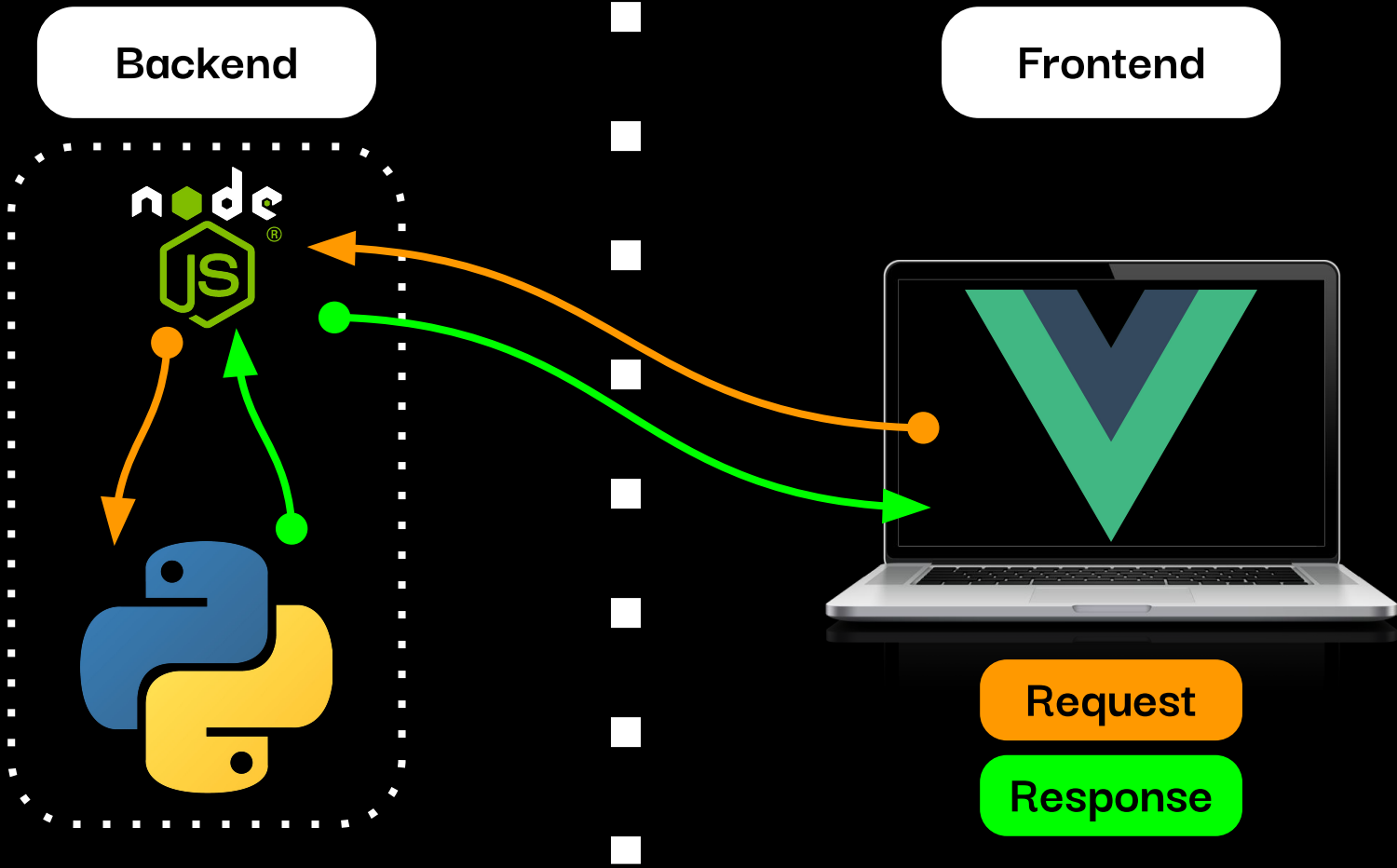
TOPIC: Develop a CNN for Image Classification using Transfer Learning

DIFFICULTY: Average

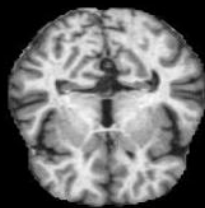
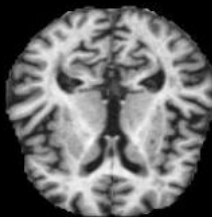
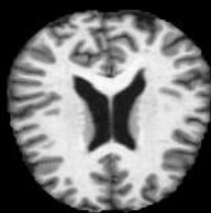
DATASET: Alzheimer Brain MRI's

MODELS: EfficientNet B4, ResNet152, DenseNet-169

ARCHITECTURE

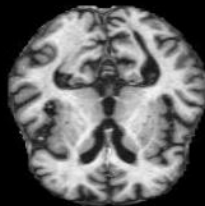
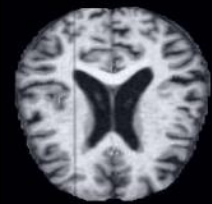


OUR DATASET



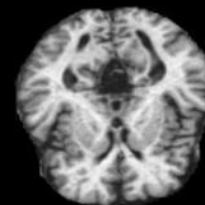
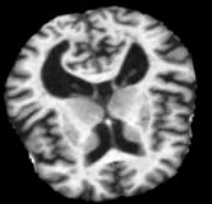
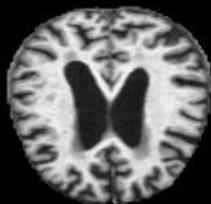
NON

DEMENTED



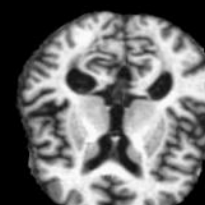
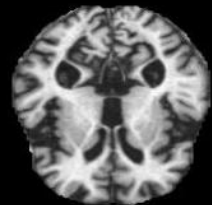
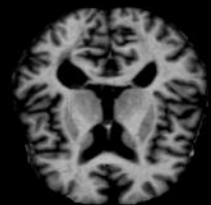
MILDLY

DEMENTED



MODERATELY

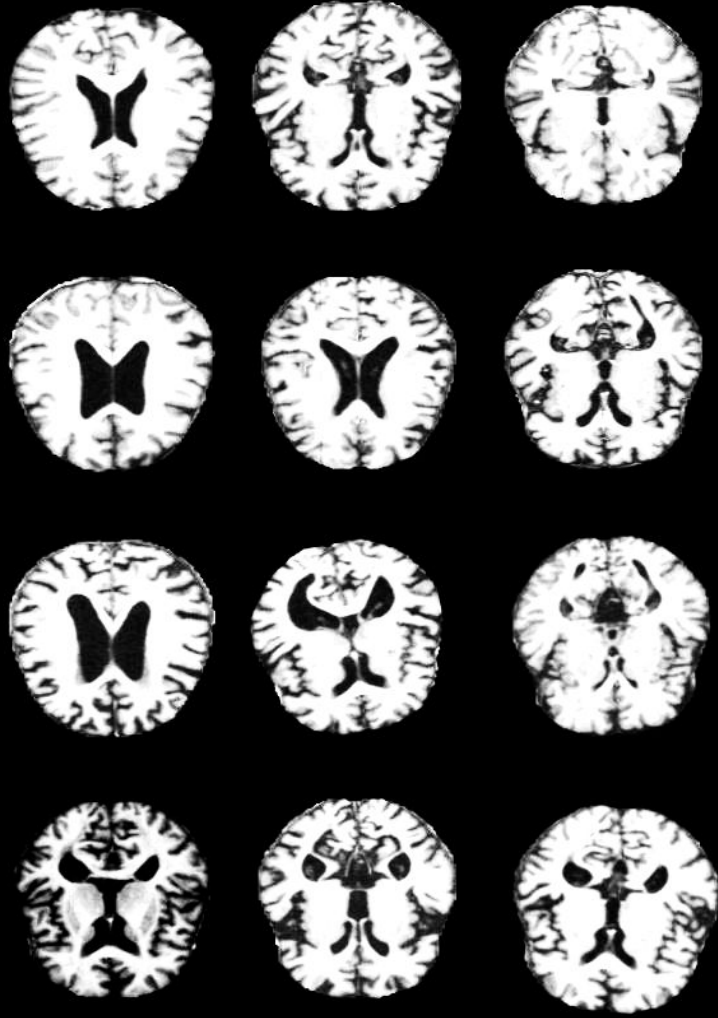
DEMENTED



SEVERELY

DEMENTED

NORMALIZATION

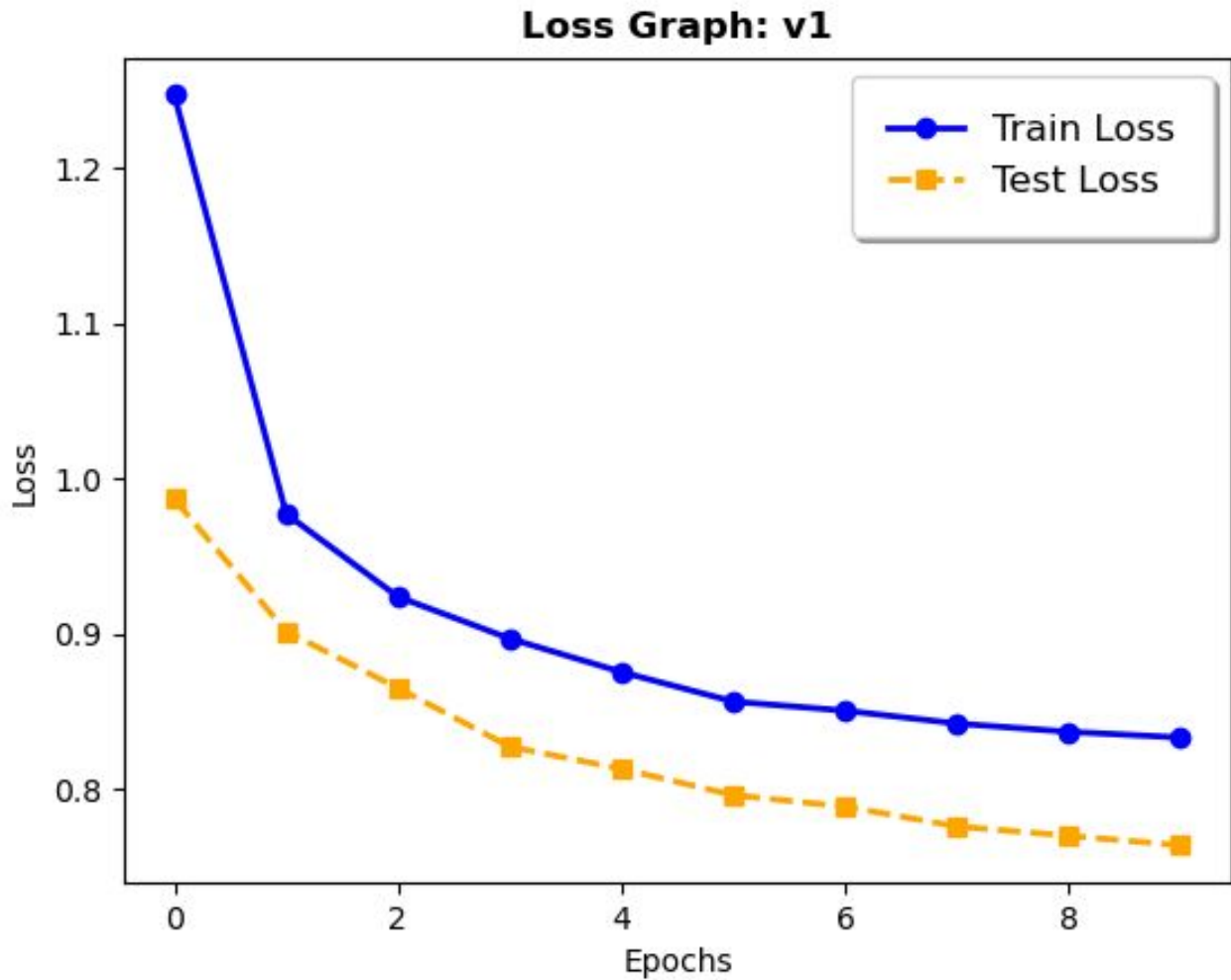


MODEL 1

EFFICIENTNET

67.16%

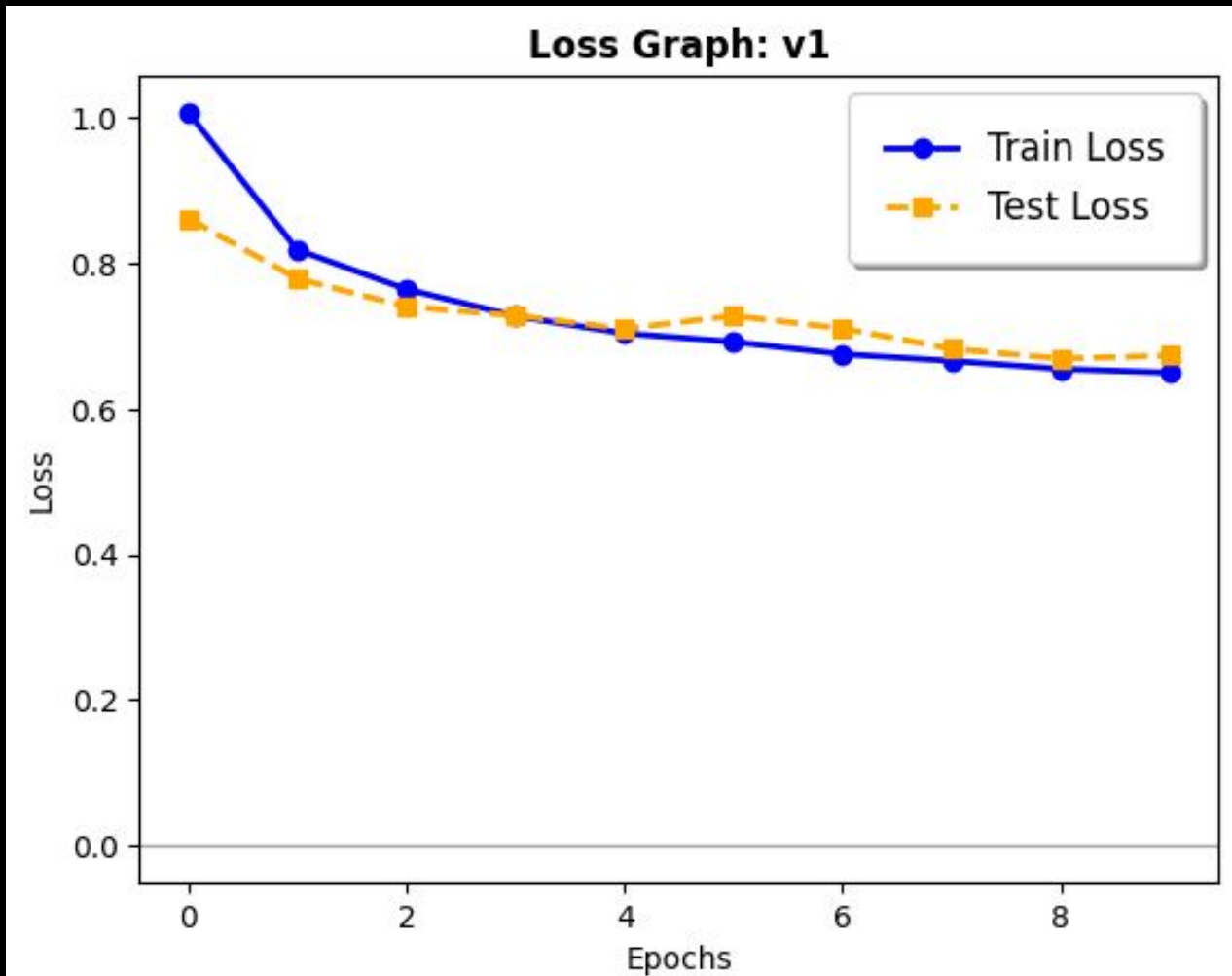
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EFFICIENTNET

PARAM. TRAINING = FALSE + SCHEDULER + NO
DROPOUT (RTX 3050 / T4 Colab)

70.49%

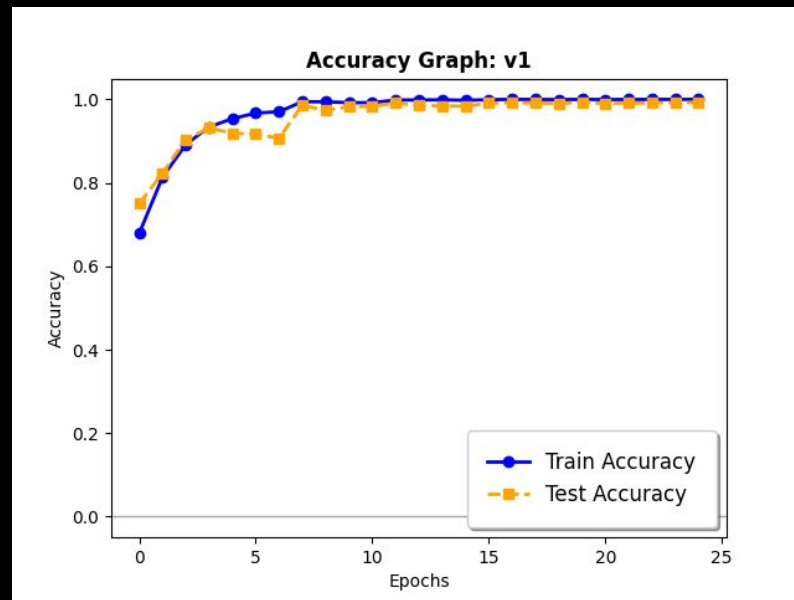
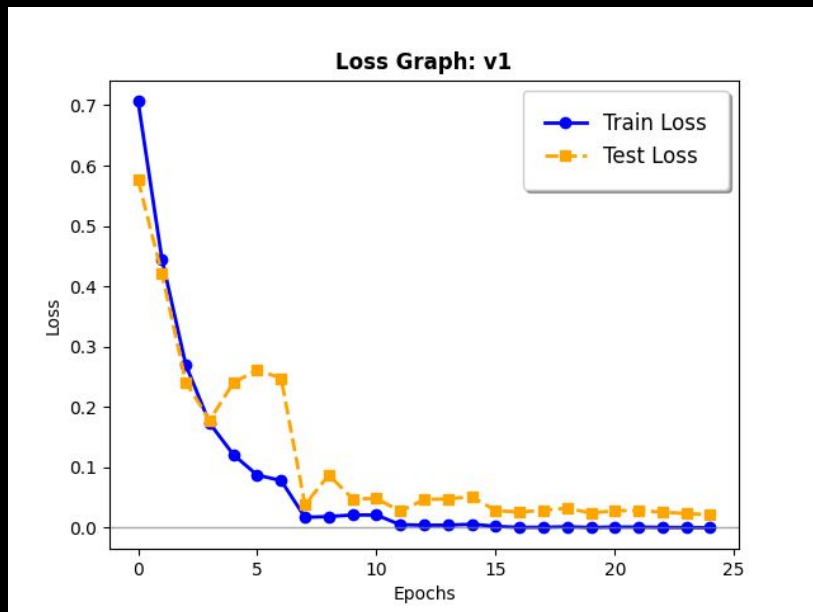


MODEL 2

RESNET152

99.31%

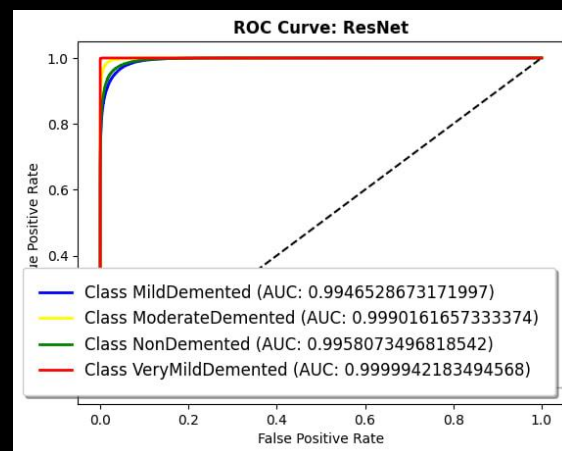
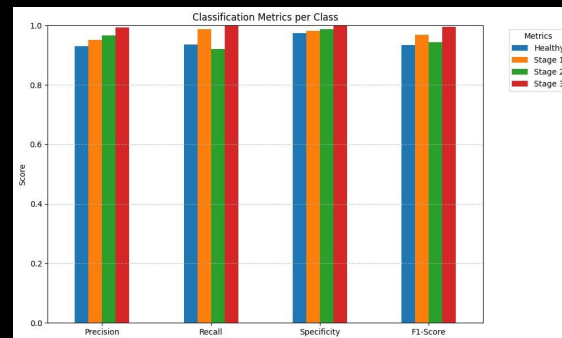
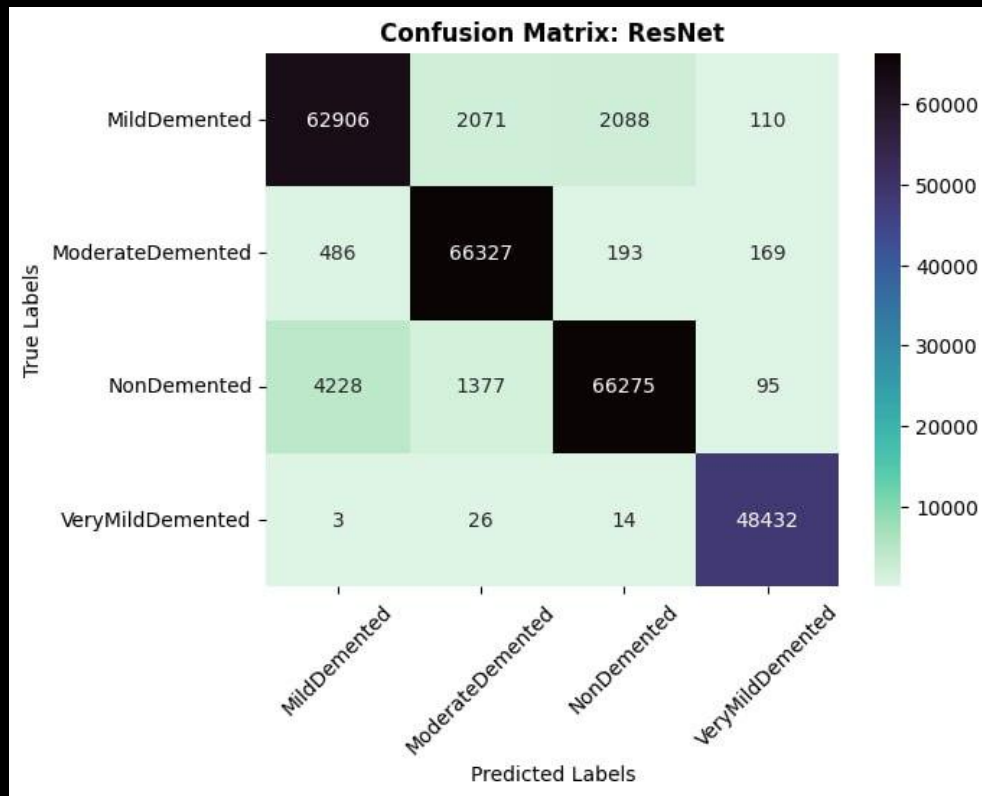
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RESNET152

99.31%

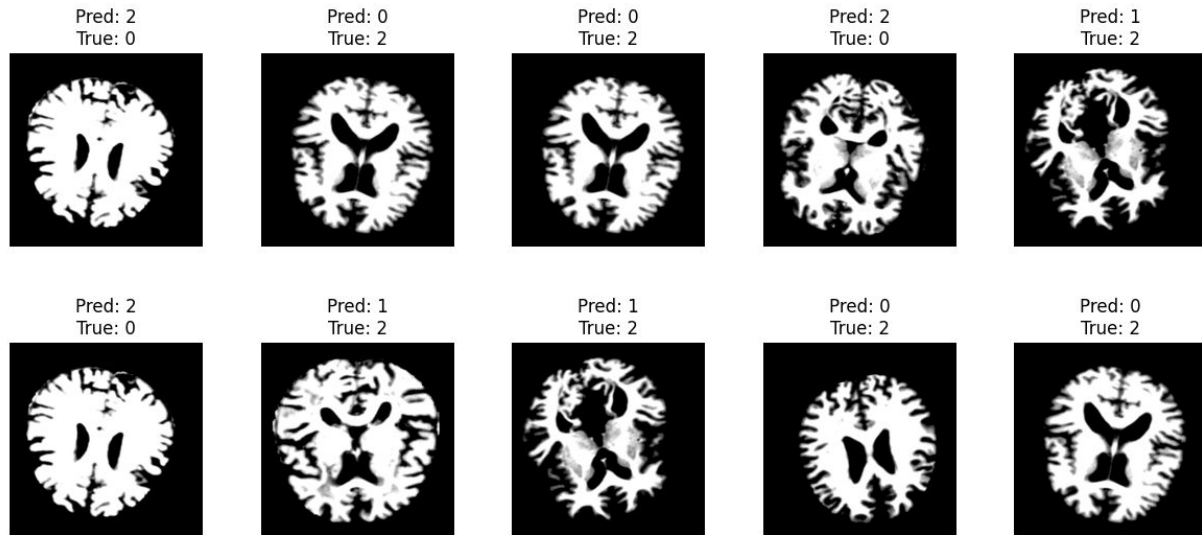
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(Vast.AI RTX 4090)



RESNET152

PARAM. TRAINING = TRUE + SCHEDULER
(Vast.AI RTX 4090)

99.31%

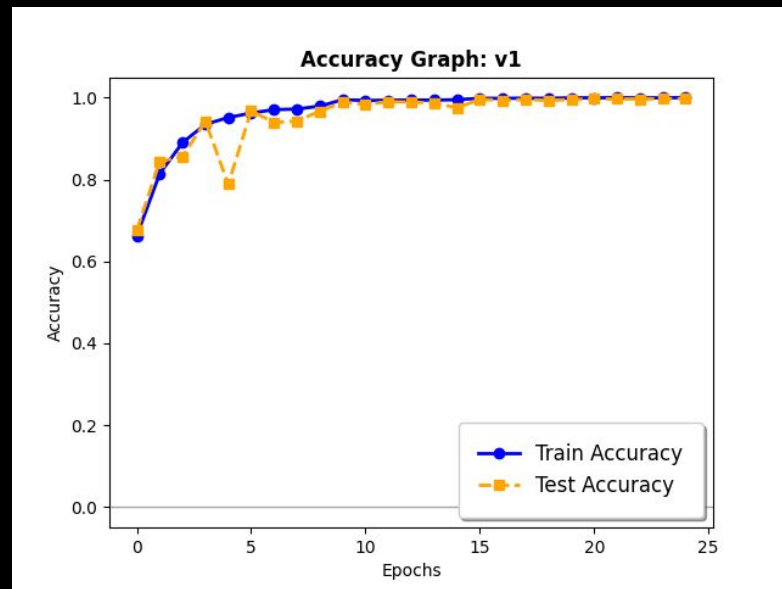
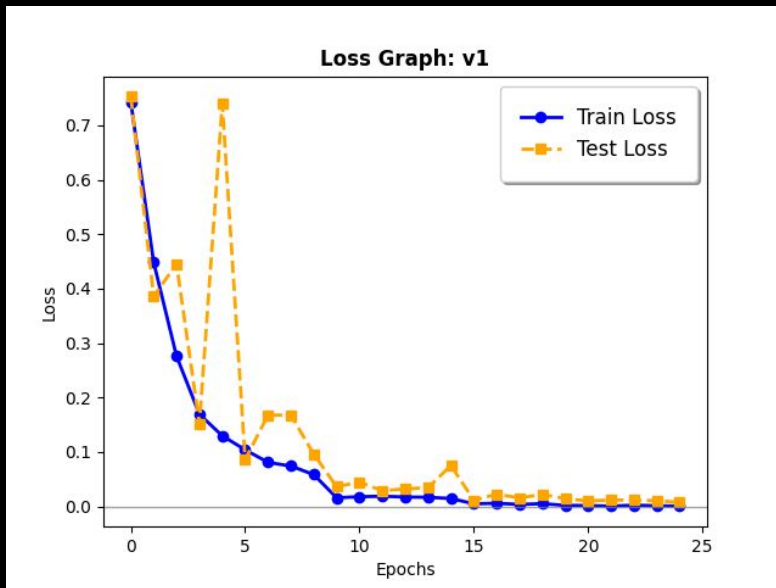


MODEL 3

DENSENET-169

99.69%

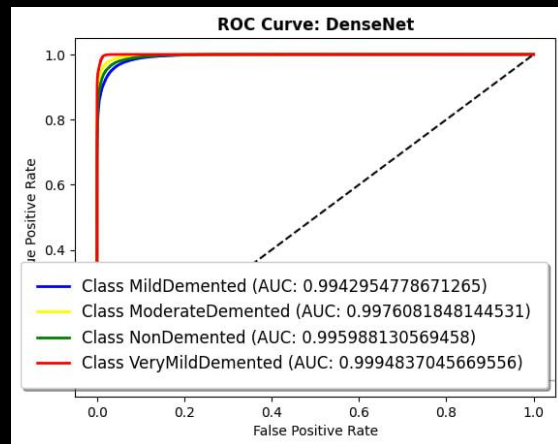
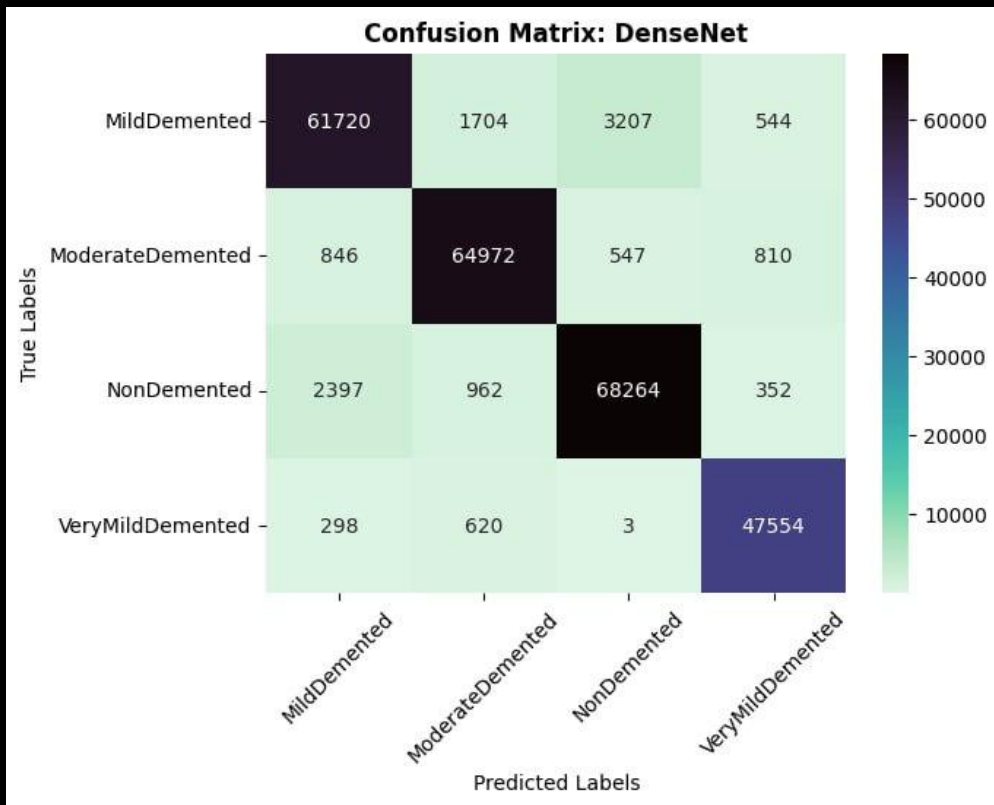
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(Vast.AI RTX 4090)



DENSENET-169

PARAM. TRAINING = TRUE + SCHEDULER
(Vast.AI RTX 4090)

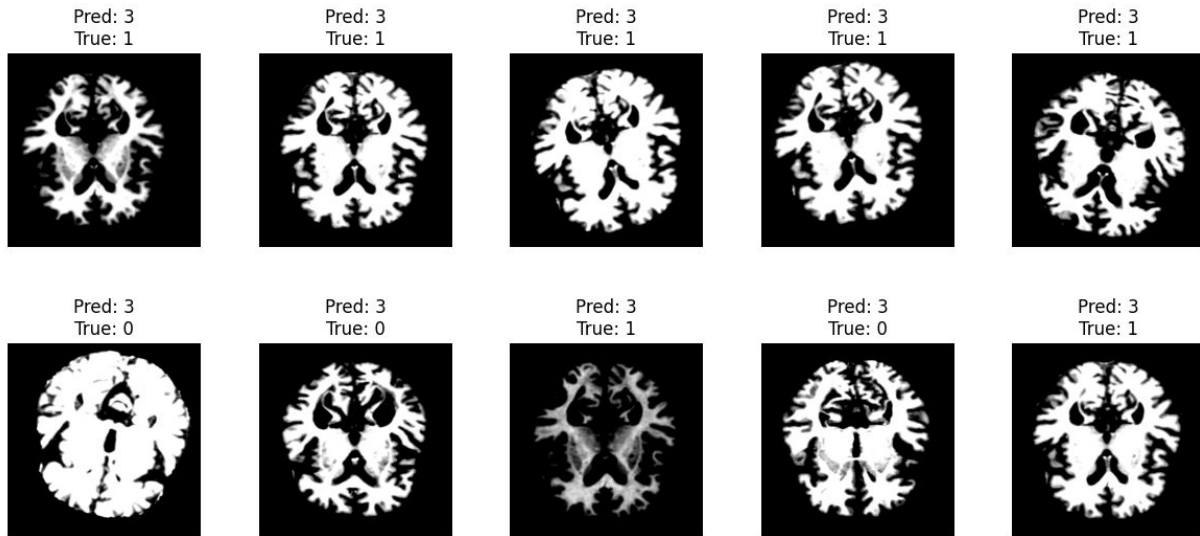
99.69%



DENSENET-169

PARAM. TRAINING = TRUE + SCHEDULER
(Vast.AI RTX 4090)

99.69%

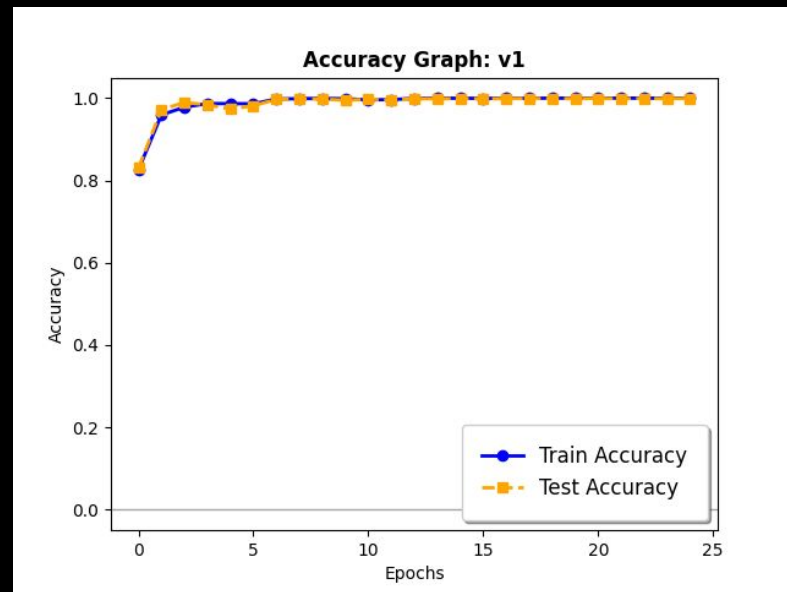
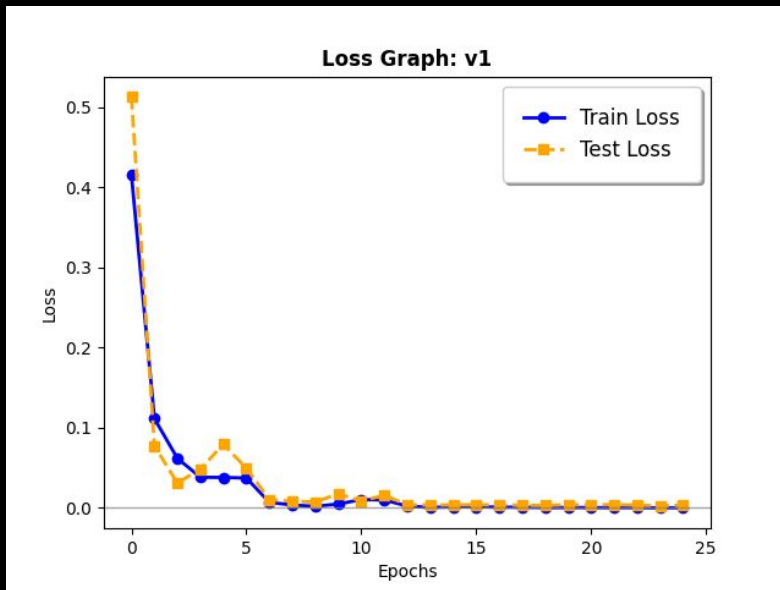


MODEL 4

EFFICIENTNET

99.91%

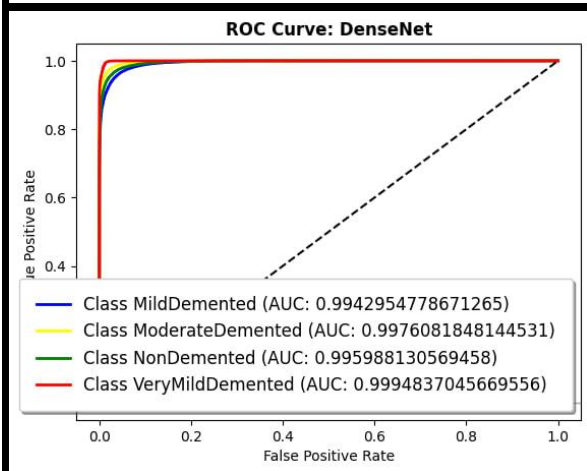
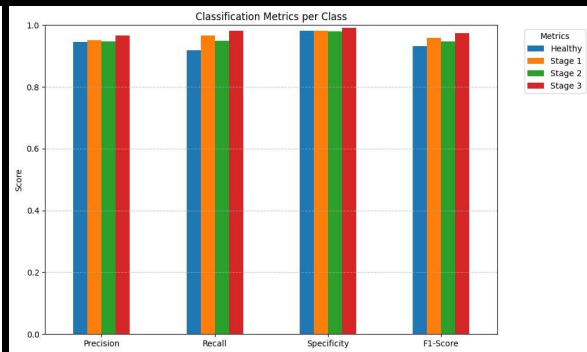
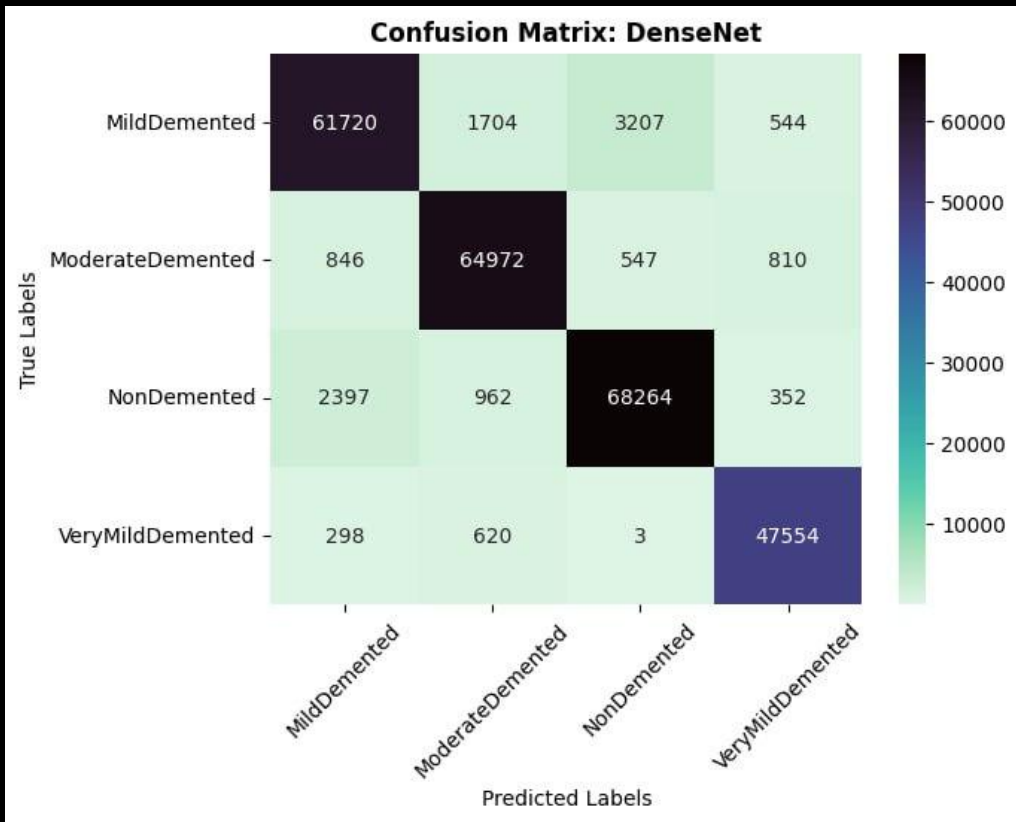
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(Vast.AI RTX 4090)



EFFICIENTNET

99.91%

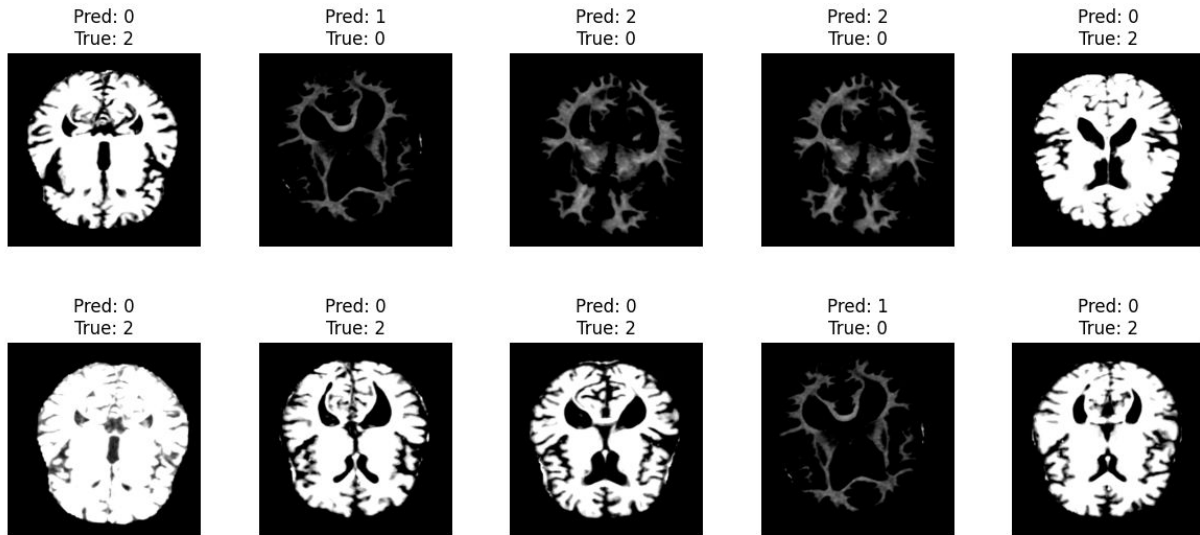
PARAM. TRAINING = TRUE + SCHEDULER
(Vast.AI RTX 4090)



EFFICIENTNET

99.91%

PARAM. TRAINING = TRUE + SCHEDULER
(Vast.AI RTX 4090)



THANK YOU !